

Playbook: reproduce every result, zero API calls

This runbook regenerates every table in the paper from the shipped caches and code. No credentials, no network. Measured wall-clock on one core (CPython 3.9, macOS): the full pipeline below completes in well under 15 minutes.

0. Environment (once)

```
$ python3 -m pip install -r requirements.txt
```

numpy, scikit-learn, matplotlib, pydantic, PyYAML, jsonschema, regex, pytest. Nothing else is needed for zero-API reproduction.

```
$ python3 -m py_compile "$(find . -name '*.py' -not -path './agentdojo_clone/*')"
```

Sanity gate: every shipped file compiles. VERIFIED.

1. Fastest end-to-end check (~2 minutes)

```
$ python3 run_taxonomy_circularity.py
```

~40 s incl. bench rebuild. VERIFIED: prints 133/440 signature-tripping draws, fused detection on the 307 evading draws = 0.2638 vs text 0.2508, tiers T.A 0.2647 / T.B 0.6316 / T.C 0.0; rewrites results/run_taxonomy_circularity.json with identical values.

```
$ python3 -m pytest tests/ -q
```

~30 s. VERIFIED: 66 passed.

2. Core paper tables (~5 minutes)

```
$ python3 run_ensemble_ablation.py
```

~48 s. VERIFIED: ablation grid (7 subsets x 3 benches), learned LR fusion, and transfer - final lines read toolcall->injecagent LR AUC = 0.9766, injecagent->toolcall 0.8447.

```
$ python3 run_fusion_rule_ablation.py
```

Fusion-rule comparison (max / mean / noisy-OR / product / LR out-of-fold) with bootstrap CIs; regenerates Table IV (section VI-E).

```
$ python3 run_diversity_stats.py
```

~15 s. VERIFIED: Kuncheva Q / disagreement / double-fault per bench and backbone-pair Spearman correlations (0.25-0.79).

```
$ python3 run_review_response.py
```

Composition-ceiling replication across 4 classifier pairings + fusion rules; regenerates section VI-G numbers.

```
$ python3 run_cost_table.py
```

~1 min. VERIFIED: Table VIII provenance - parses wall-clock and call counts from run_adaptive_fusion_eval_glm.log, recomputes prompt length (2,098 chars over the 440 attacks) and pattern latency, and re-derives costs (\$0.36 text GLM-4.6 / \$1.63 CaMeL est / \$1.99 fused per 1,000 attack trajectories).

Playbook page 2: statistics, figures, documents

3. Statistical treatment (a few minutes)

\$ python3 run_ablation_ci.py

Full-grid 2,000-replicate paired bootstrap CIs for AUC and TPR@7%FPR on every subset x bench cell, all 21 pairwise McNemar tests per bench with Holm correction, and the AgentDojo pattern-channel sign-stability check (expect: all replicates below AUC 0.5, CI [0.111, 0.195]). CPU-bound; the slowest script in the artifact.

\$ python3 run_contamination_prepost.py --help

Pre-fix vs post-fix construction uses the same loader with contamination_fix=False; the cache glm_guard_injecagent_prefix.jsonl holds the pre-fix GLM-4.6 scores, so the comparison also runs without API calls (see script source for the exact flag).

\$ python3 run_paired_head_diffs.py

~1 min. VERIFIED: paired AUC-difference CIs of the learned head and the text+provenance subset against the text channel alone - the head matches text on InjecAgent, diff CI [-0.0041, +0.0127]; adds on tool-call, CI [0.0183, 0.0594]; taxonomy-independent subset gain CI [0.0053, 0.0233].

\$ python3 run_tpr_at_1pct.py

Re-evaluates the ablation grid at 1% FPR and reports the score-granularity diagnostics: the text guard emits 18 distinct scores and the benign tie plateau at 0.15 makes TPR@1% = TPR@7% for every subset (actual FPR 0.42%).

\$ python3 run_adaptive_mcnemar.py

~10 s. VERIFIED: adaptive-bench McNemar at matched 7%-FPR thresholds from the shipped caches - b = 0 in all three comparisons (c = 33, 37, 4): disagreements are one-directional.

4. Figures

\$ cd figs && python3 make_figs.py && python3 make_figs2.py

Regenerates the five published figures (regime heatmap, ablation heatmap, adaptive AUC bars, per-shape detection, contamination pre/post) as PDFs in figs/.

\$ python3 make_RESULTS.py

Rebuilds RESULTS.pdf: every published number rendered as tables read live from results/*.json, plus all five figures.

5. Documents

\$ python3 make_README.py && python3 make_PLAYBOOK.py

Rebuild README.pdf and this runbook.

6. Expected-values checklist

After running steps 1-3, these exact values must appear (any deviation means the environment altered numerics):

\$ ablation grid, tool-call triple: AUC 0.874

text alone 0.832; text+pattern 0.861; +15.4 pp TPR@7%FPR (35.4 -> 50.8).

\$ adaptive bench, fused: AUC 0.795 [0.766, 0.825]

text 0.778; paired-difference CI [0.011, 0.026]; McNemar b=0, c=37.

\$ circularity split: 133 trip / 307 evade

evading-draw detection 0.2638 vs text 0.2508; T.C (data-flow) 0.0.

\$ contamination: pre AUC 0.518 / 81.4% FPR -> post 0.916 / 0%

same classifier GLM-4.6, 510 records, only the 323 benign rows differ.

Playbook page 3: optional live re-scoring, troubleshooting

None of the following is required to reproduce any published number; it re-scores trajectories against live endpoints and will differ in transport noise.

7. Optional: re-score the text channel (OpenRouter)

```
$ export OPENROUTER_API_KEY=... # never commit this
```

Provider for the static benches (Claude Haiku 4.5 text channel, CaMeL P/Q, comparators).

```
$ python3 run_or_extra_comparison.py --dry-run 5
```

Every API-touching script ships a small-N dry-run flag; use it before any full run. Caches append to results/cache/, so a re-run resumes rather than double-spends.

8. Optional: backbone deconfound (any OpenAI-compatible endpoint)

```
$ export GATEWAY_API_KEY=... GATEWAY_BASE_URL=.../v1
```

Re-scores ONLY the provenance channel (CaMeL P/Q) on a second backbone (default google/gemma-4-31b-it), then recomputes the tool-call fused/subset numbers from caches. The gateway must accept x-api-key header auth; 429s back off automatically and failed calls are never cached as verdicts. ~1,150 calls total.

```
$ python3 run_backbone_deconfound.py --dry-run 8 --workers 2
```

Probe + 8 trajectories (~20 calls) with transport-health gate before the full run; cache resumes on rerun.

```
$ python3 run_backbone_deconfound.py --bench injecagent
```

Off-regime companion (P2 backbone-independence): re-scores provenance on the 510 InjecAgent trajectories. Measured cost: 103 calls - the CaMeL policy short-circuits trajectories with no untrusted-to-sensitive flow before any LLM call. Same env, same cache-resume behavior; --analyze-only reruns both benches from caches with zero calls.

9. Optional: adaptive bench (z.ai / GLM-4.6)

```
$ export ANTHROPIC_BASE_URL=https://api.z.ai/api/anthropic
```

WARNING: this endpoint silently aliases claude- model ids to GLM models. The script probes the endpoint first and labels itself GLM-4.6; treat any claude-* id on this transport as GLM.*

```
$ python3 run_adaptive_fusion_eval.py --provider glm --dry-run 10
```

~30 calls. Full run: 440 attacks + 240 benign, ~2,800 calls, documented abort gates (transport-empty 20%, probe mismatch). --provider anthropic exists for a directly-funded Anthropic run.

10. Troubleshooting

MODULENOTFOUNDERROR: OSI_BENCH / BENCHMARKS / ADAPTIVE

AGENTDOJO RUNS NOT FOUND AT AGENTDOJO_CLONE/RUNS/...

INJECAGENT LOADER PATH

SKLEARN RUNTIMEWARNINGS IN MATMUL

PYTEST COLLECTS 66 TESTS